Project Planning & Management Document

1. Project Overview

Project Title *– Aether Labs: The Final Override (Interactive Smart Escape Room Model)*

Team Members & Roles –
Himani Sinojiya – Coding, Documentation, Physical model building, Interfacing, Mechanisms
Swara Zore – Coding, Documentation, Physical model building, Interfacing, Mechanisms

Aether Labs: The Final Override *is a miniature interactive escape room model based on a futuristic laboratory storyline. The model contains three rooms representing three security levels, where players solve physical puzzles using electronics, sensors, lights, sound effects, and mechanical mechanisms to progress through each stage and finally detonate the invention to complete the mission.*

2. Team Roles & Responsibilities

Member Name	Role	Responsibilities
Himani Sinojiya	Project Co-Developer	Shared responsibility in planning, coding, assembly (model), debugging, documentation, interfacing, puzzle development, testing, and final presentation
Swara Zore	Project Co-Developer	Shared responsibility in planning, model making, fabrication, electronics setup, puzzle development, interfacing, coding, testing, and final presentation

Since this was a two-member project, most tasks were carried out collaboratively, with both members contributing across planning, fabrication, electronics, testing, and final execution.

3. Project Timeline & Milestones

Week	Goals	Deliverables	Status
Week 0 (Mar 30 – Apr 4)	Finalise project concept, storyline, and game theme	Final idea of sci-fi escape room with Aether Labs storyline and 3 security levels, webpage building, puzzles development	Completed (with some alterations)
Week 1 (Apr 5 – Apr 11)	Design structure and begin model fabrication	Base layout of three-room escape model, room partitions, initial material cutting, and physical model building	Completed
Week 2 (Apr 12 - Apr 18)	Electronics integration, puzzle setup, testing and coding	Trial run of individual electronic components and interfacing them on the model	Completed
Week 3 (Apr 19 – Apr 22)	Final finishing, testing and video of the final model	Debugging, final presentation-ready model, video	Partially completed

Break project into weekly achievable goals

4. Weekly Progress Log

A table for each week:

The project followed an iterative build-test-improve cycle. Each week involved practical problem-solving, adapting plans based on time constraints, and improving the model through continuous testing.

Week 0 (Mar 30 – Apr 4)

Planned Tasks	*Brainstorm and ideate, finalize project concept, develop storyline, decide game mechanics and puzzle flow*
Completed Tasks	*Selected sci-fi escape room concept, created Aether Labs storyline, finalized three security level structures, and brainstormed puzzles*
Challenges Faced	*Multiple ideas created confusion, and the initial web-based concept of an escape room was too ambitious for the timeline*
Solutions	*Prioritised feasible features, shifted focus to a stronger physical interactive model with electronics*
Next Steps	*Begin physical structure design and collect materials*

👉 Encourage honesty here—failures are important learning

Week 1 (Apr 5 – Apr 11)

Planned Tasks	*Design room structure, build base model, create partitions, begin fabrication*
Completed Tasks	*Constructed base layout of three-room model, completed walls and room divisions, started external design elements*
Challenges Faced	*Difficulty maintaining measurements, alignment, and model stability*
Solutions	*Re-measured dimensions, reinforced joints, adjusted room spacing*
Next Steps	*Continue detailing and begin integrating electronic components*

👉 *Encourage honesty here—failures are important learning*

Week 2 (Apr 12 – Apr 18)

Planned Tasks	*Integrate electronics, develop final code logic for puzzle flow, test sensors and outputs, connect components to the model*
Completed Tasks	*Completed trial runs of LEDs, buzzers, sensors, switches, and basic interfacing. Developed initial code for room progression, puzzle triggers, and security level flow. Tested multiple puzzle sequences on the model*
Challenges Faced	*The first versions of the code did not follow the intended game flow consistently. Some sensors triggered incorrectly, delays caused sequence errors, and multiple components working together created errors. Hardware responses sometimes failed after repeated trials.*
Solutions	*Debugged the program in stages by testing one component at a time, then combining modules gradually. Rewrote sections of the code several times, simplified logic conditions, adjusted timings, and reorganised the flow for smoother progression through all three rooms. Improved wiring stability during code testing*
Next Steps	*Final assembly, full gameplay testing, optimise reliability, add aesthetic finishing, and prepare final presentation model.*

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Week 3 (Apr 19 – Apr 22)

Planned Tasks	*Final finishing, complete testing, record project video, prepare presentation, and ensure all puzzles run smoothly*
Completed Tasks	*Major debugging completed, visual finishing added, documentation prepared, and video work in progress. Final testing of the room progression and puzzle flow was carried out*
Challenges Faced	*Loose wire connections repeatedly interrupted circuits, causing components to stop working unexpectedly during testing. Some outputs worked inconsistently when the model was moved. Time pressure and last-minute technical issues added difficulty*
Solutions	*Rechecked all wiring connections, tightened loose joints, reorganised cables for better stability, and retested each circuit individually before running the full system. Prioritised fixing critical electrical issues first and divided the remaining tasks efficiently between both members*
Next Steps	*Final demo rehearsal, complete video submission, ensure stable setup during presentation*

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